

MERCED CASTLE, CA, USA

WMO: 724810

Lat: 37.383N

Lon: 120.567W

Elev: 191

StdP: 14.59

Time Zone: -8.00 (NAP)

Period: 90-19

WBAN: 23203

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	30.0	31.9	21.9	16.6	50.5	26.8	20.9	48.4	26.9	54.4	23.3	53.0	1.1	140	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.9	102.5	69.6	100.0	68.3	98.3	67.5	72.1	96.0	70.7	94.3	69.3	92.6	9.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	91.9	74.5	63.3	87.8	72.5	61.4	81.9	73.0	35.7	94.8	34.5	94.9	33.4	92.9	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	17.6	14.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	46.1	51.5	56.1	61.6	69.2	74.6	80.3	78.3	75.1	65.8	54.3	46.2
	DBStd	13.04	4.30	5.02	5.25	6.29	6.44	5.97	4.66	4.93	4.58	6.33	5.32	5.04
	HDD50	341	133	38	12	1	0	0	0	0	0	0	16	141
	HDD65	2384	586	377	277	139	31	3	0	0	1	65	322	583
	CDD50	5208	12	81	201	351	596	738	939	877	754	489	146	24
	CDD65	1774	0	0	1	38	162	291	474	412	306	88	2	0
	CDH74	24564	0	7	75	727	2376	4037	6481	5506	3995	1293	67	0
	CDH80	13751	0	0	4	259	1150	2249	4005	3298	2257	519	10	0
(14) Wind	WSAvg	5.5	3.6	4.6	5.2	6.0	7.0	6.9	7.5	6.9	5.7	4.9	3.8	3.8
(15) Precipitation	PrecAvg	12.4	2.5	2.3	1.9	0.9	0.6	0.1	0.0	0.0	0.0	0.8	1.0	2.2
	PrecMax	22.0	6.5	7.6	5.4	3.2	2.5	1.1	0.2	0.1	0.2	3.1	2.6	5.2
	PrecMin	4.3	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	4.2	1.8	1.6	1.5	0.9	0.6	0.2	0.0	0.0	0.1	0.8	0.7	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	74.8	80.7	92.7	99.6	106.3	106.4	105.8	102.0	97.0	81.3	65.9
		MCWB	53.7	58.7	60.1	65.4	67.8	70.9	71.3	69.7	68.0	67.5	63.1	54.1
	2%	DB	61.8	70.1	75.3	86.8	95.0	100.2	102.9	101.7	99.3	90.6	74.8	62.6
		MCWB	53.8	57.7	59.0	62.4	64.9	69.2	70.3	68.4	66.6	65.2	59.2	53.9
	5%	DB	58.5	65.9	72.6	82.0	91.0	96.9	100.0	99.1	96.6	85.5	71.0	59.4
		MCWB	52.1	56.2	58.9	60.7	63.8	67.0	68.8	67.6	66.1	63.8	58.3	52.3
	10%	DB	55.3	63.0	69.5	77.6	86.9	92.0	97.3	95.7	92.4	81.4	66.3	56.6
		MCWB	50.5	54.4	57.9	59.5	62.6	65.6	68.4	66.7	65.3	61.6	56.2	51.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.6	63.1	64.3	71.4	70.1	73.0	75.1	72.4	70.2	71.5	65.5	61.4
		MCDB	60.7	70.0	70.9	81.8	95.5	98.8	101.6	99.6	96.0	87.1	73.3	61.7
	2%	WB	55.3	59.5	61.9	65.3	67.6	71.2	72.6	70.8	68.8	68.5	63.1	58.0
		MCDB	57.9	65.8	70.6	79.1	91.1	95.5	96.6	96.7	93.6	83.6	69.5	59.0
	5%	WB	53.7	57.5	60.4	63.1	65.3	69.3	70.9	69.2	67.6	66.1	61.2	55.3
		MCDB	56.6	63.7	69.7	76.7	86.8	90.5	94.8	93.4	91.5	81.1	67.6	57.1
	10%	WB	51.8	55.5	58.8	61.0	63.3	67.7	69.4	67.9	66.4	64.0	58.9	53.7
		MCDB	54.8	61.8	67.6	75.3	83.4	89.5	94.0	92.2	89.1	76.1	64.9	54.8
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	21.4	23.0	27.8	30.6	32.1	33.9	33.7	34.1	29.5	25.3	19.4
		MCDBR	24.7	27.4	30.0	34.7	36.1	37.2	36.1	36.7	39.3	35.9	31.1	23.6
	5% WB	MCWBR	15.7	16.4	15.1	14.0	12.7	12.5	12.6	12.4	12.3	14.1	16.0	15.3
		MCWBR	16.0	22.9	25.4	31.2	32.9	34.0	33.0	33.9	35.1	31.0	25.0	15.5
(40) Clear-Sky Solar Irradiance	taub		0.347	0.349	0.356	0.358	0.362	0.352	0.366	0.352	0.354	0.362	0.351	0.330
		taud	2.384	2.403	2.382	2.388	2.398	2.434	2.409	2.424	2.430	2.395	2.389	2.462
	Ebn at Noon		263	279	288	292	292	293	288	291	285	270	259	262
		Edh at Noon	29	32	36	37	37	36	37	36	33	31	28	25
(44) All-Sky Solar Radiation	RadAvg		738	1085	1555	1994	2388	2658	2590	2347	1929	1398	927	676
	RadStd		120	121	142	161	122	84	78	65	61	81	56	99

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (44 neighbors)		+0.72	N/A	-2.07	N/A	N/A	N/A	N/A	-205	+248	+63	

Nomenclature: See separate page